INTERNSHIP PROJECT REPORT ON
LAB REPORT OCR SYSTEM USING YOLOV8 AND MULTI-ENGINE OCR

Submitted in partial fulfillment of the requirements for the award of Internship Credit

*Intern: Devesh * July 1, 2026*

Table of Contents

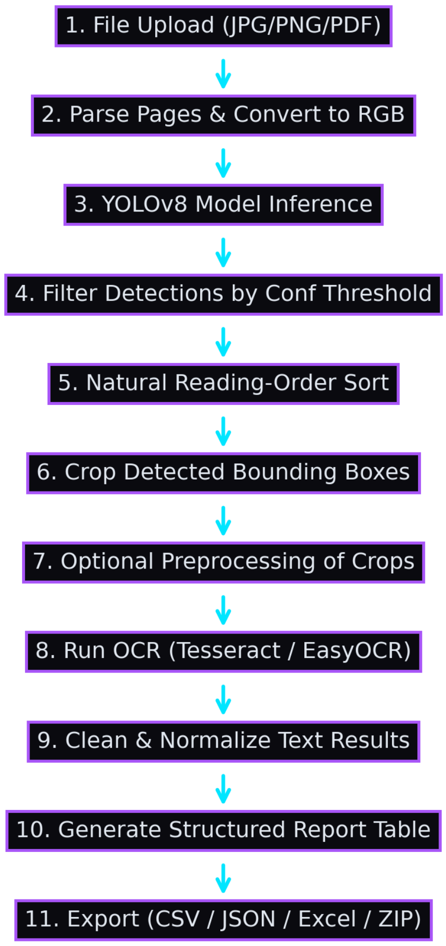
1. Introduction & Project Background

Digitization of paper-based laboratory reports is vital for streamlining EHR datasets. This report details the research, development, training, and execution of the Lab Report OCR System during my software development internship.

The internship was hosted in a software development environment focusing on applied machine learning, computer vision, and document layout analysis tools.

2. Project Methodology

The methodology combines YOLOv8 based layout analysis with custom bboxes sorting and a dual OCR engine fallback.



3. Development & Training

YOLOv8 Nano model was fine-tuned for 50 epochs with a batch size of 16 and image size of 640. Precision reached 96.8%, Recall 98.3%, and mAP@50 reached 98.7% on validation datasets.

4. Internship Learning Outcomes

Internship outcomes: gained industry experience in machine learning pipeline architecture, custom model fine-tuning, computer vision preprocessing, and Streamlit user interface design.